DATA 8

Spring 2024

Lecture 25

Center and Spread

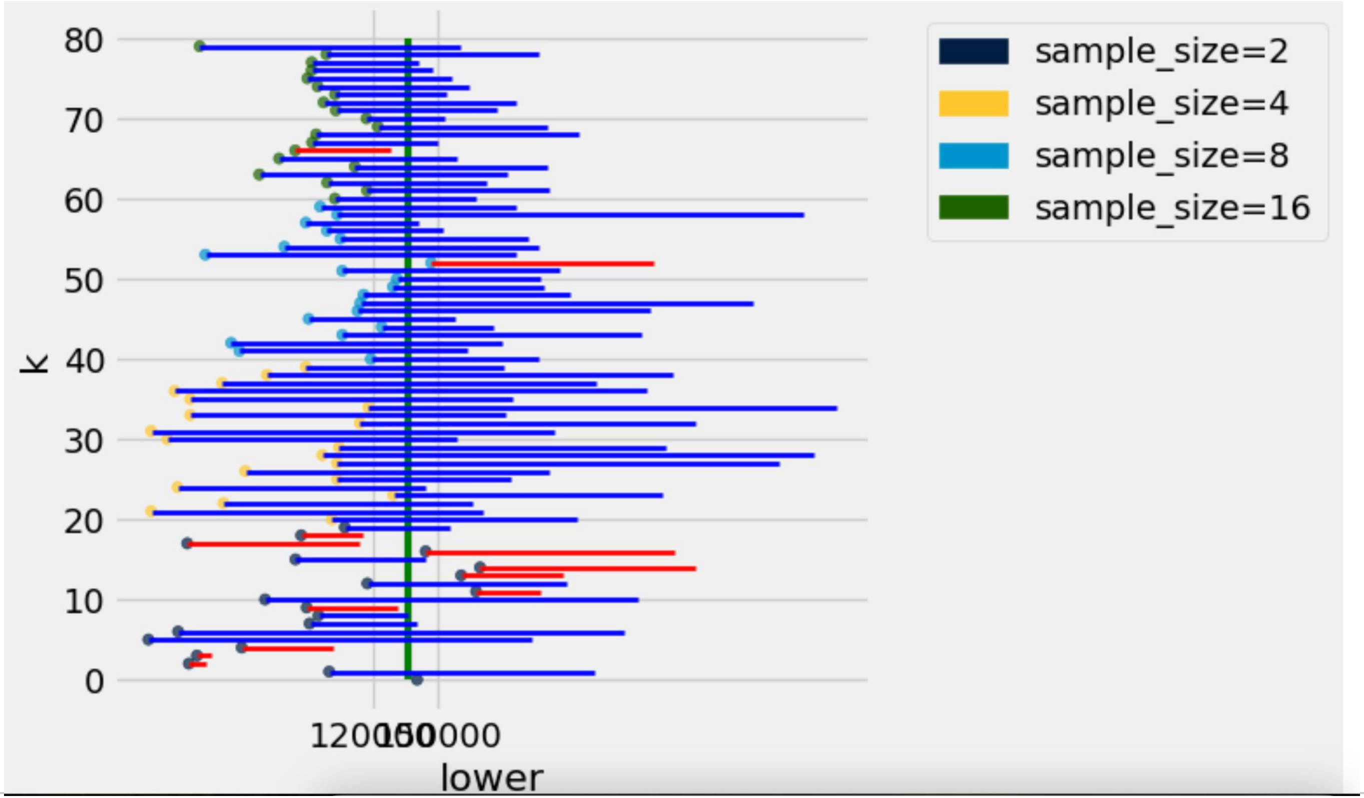
Announcements

- **Homework 8** is due Wednesday 3/20
- **Lab 7** is due today at 5pm
- Midterm grades + grade report will hopefully be released today!
  - Waiting for some homework grading to wrap-up
- **Tutoring section sign-ups** release this Friday (3/15) evening around 6 pm

Recap of Last Lecture

- Confidence Interval
  - Interval of estimates of a parameter
  - Based on random sampling
  - The process results in a random interval
  - A “good” interval is one that contains the parameter
  - The **confidence is in the process** that creates the interval:
    - It generates a “good” interval with chance 95%
- Bootstrap sample size: Same as the random sample
  - Smaller sample size yield uncertain estimates

Confidence Intervals vs Sample Size



When *Not* to Use Our Bootstrap Method

- If you're trying to estimate any parameter that's greatly affected by rare elements of the population
  - Very high or very low percentiles, or min and max
- If the probability distribution of your statistic is not roughly bell shaped (the shape of the empirical distribution will be a clue)
- If the original sample is very small

Can You Use a CI Like This?

By our calculation, an approximate 95% confidence interval for the average age of the mothers in the population is (26.9, 27.6) years.

True or False:

- About 95% of the mothers in the population were between 26.9 years and 27.6 years old.

Answer: False.

- We are estimating average age
- There is a 95% chance that 95%CI will cover the true value of the parameter

Is This What a CI Means?

An approximate 95% confidence interval for the average age of the mothers in the population is (26.9, 27.6) years.

True or False:

- There is a 0.95 probability that the average age of mothers in the population is in the range 26.9 to 27.6 years.

Answer: False. The parameter is fixed, and the interval (26.9, 27.2) is fixed. The parameter is either in that interval, or not. Once you've picked an interval, there's no probability involved.

(CI is a statement about the fraction of intervals containing the parameter not an individual interval)

Confidence Intervals For Testing

Using a CI for Testing

- **Null hypothesis: Population average = *x***
- **Alternative hypothesis: Population average ≠ *x***
- **Cutoff for p-value**: *p*%
- Method:
  - Construct a (100-*p*)% confidence interval for the population average
  - If *x* is not in the interval, reject the null
  - If *x* is in the interval, can't reject the null

Center and Spread

Questions

- How can we quantify natural concepts like “center” and “variability”?
- Why do many of the empirical distributions that we generate come out bell shaped?
- How is sample size related to the accuracy of an estimate?

Average

The Average (or Mean)

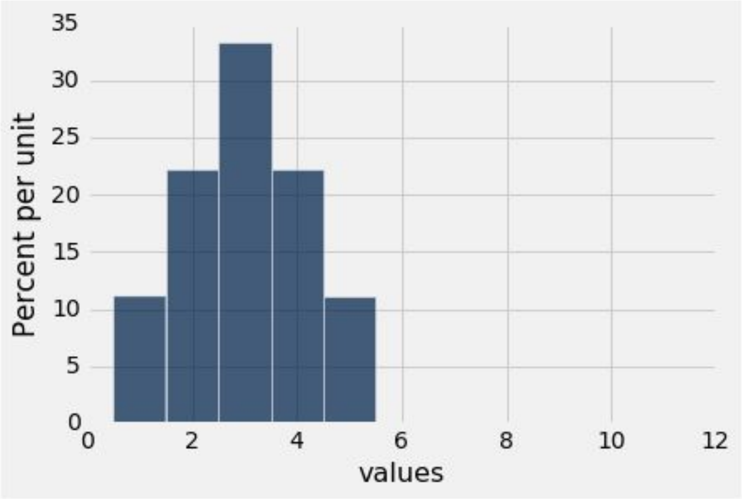
Data: 2, 3, 3, 9 **Average = (2+3+3+9)/4 = 4.25**

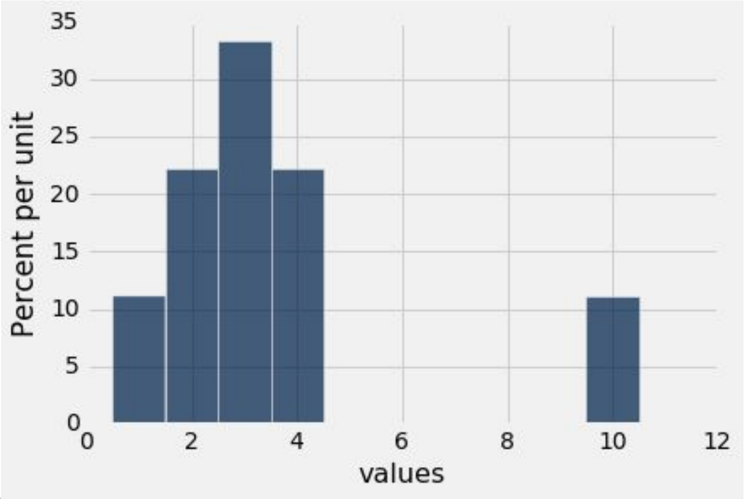
- Need not be a value in the collection
- Need not be an integer, even if the data are integers
- Somewhere between `min` and `max`, but not necessarily halfway in between
- Same units as the data
- Smoothing operator: collect all the contributions in one big pot, then split evenly

(Demo)

Discussion Question

Are the medians of these two distributions the same or different? Are the means the same or different? If you say "different," then say which one is bigger.

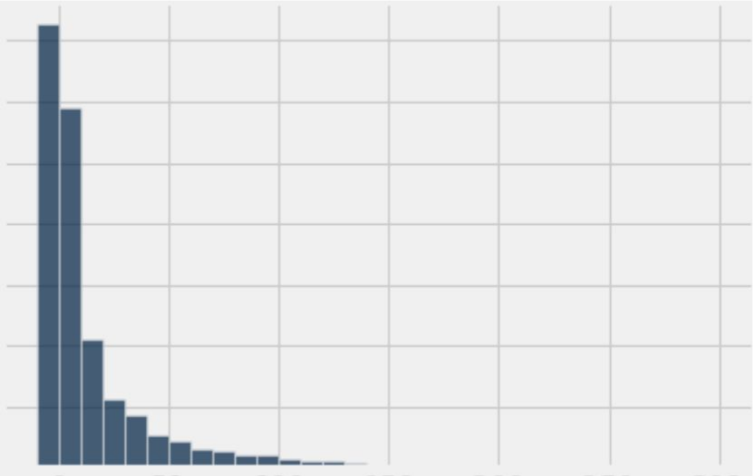




Comparing Mean and Median

- **Mean:** Balance point of the histogram
  - Physics Analogy: Center of Gravity
- **Median:** 50th percentile of the data
- If the distribution is symmetric about a value, then that value is both the average and the median
- If the histogram is skewed, then the mean is pulled away from the median in the direction of the skew (tail)

Discussion Question

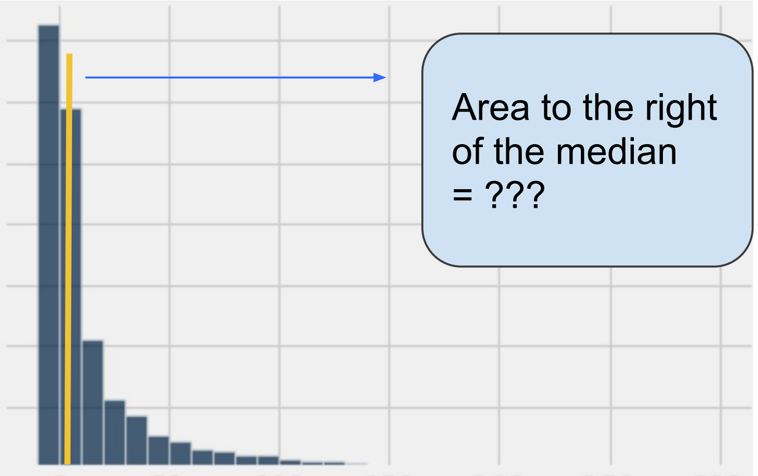


The histogram shows the distribution of values contained in an array **`x`**.

Which of the following is **`True`**?

```
(A)  sum(x > np.average(x)) / len(x) < 0.5
(B)  sum(x > np.average(x)) / len(x) == 0.5
(C)  sum(x > np.average(x)) / len(x) > 0.5
```

Hint



The histogram shows the distribution of values contained in an array **`x`**.

The gold line is at the median.

Which of the following is **`True`**?

```
(A)  sum(x > np.average(x)) / len(x) < 0.5
(B)  sum(x > np.average(x)) / len(x) == 0.5
(C)  sum(x > np.average(x)) / len(x) > 0.5
```

Standard Deviation

Defining Variability

Plan A: “biggest value - smallest value”

- Doesn’t tell us much about the shape of the distribution

Plan B:

- Measure variability around the mean
- Need to figure out a way to quantify this

(Demo)

How Far from the Average?

- Standard deviation (SD) measures roughly how far the data are from their average

- SD = Root Mean Square of Deviations from Average

  Steps: 5 ← 4 ← 3 ← 2 ← 1

  (SD is known as the RMS of the deviations)

- SD has the same units as the data

Why Use the SD?

There are two main reasons.

- **The first reason:**

No matter what the shape of the distribution,
the bulk of the data are in the range “average ± a few SDs”

- **The second reason:**

Coming up next time.

Chebyshev's Inequality

How Big are Most of the Values?

No matter what the shape of the distribution,

the bulk of the data are in the range “average ± a few SDs”

Chebyshev’s Inequality

No matter what the shape of the distribution,

the proportion of values in the range “mean ± z SDs” is

at least $1 - 1/z^2$

Chebyshev's Bounds

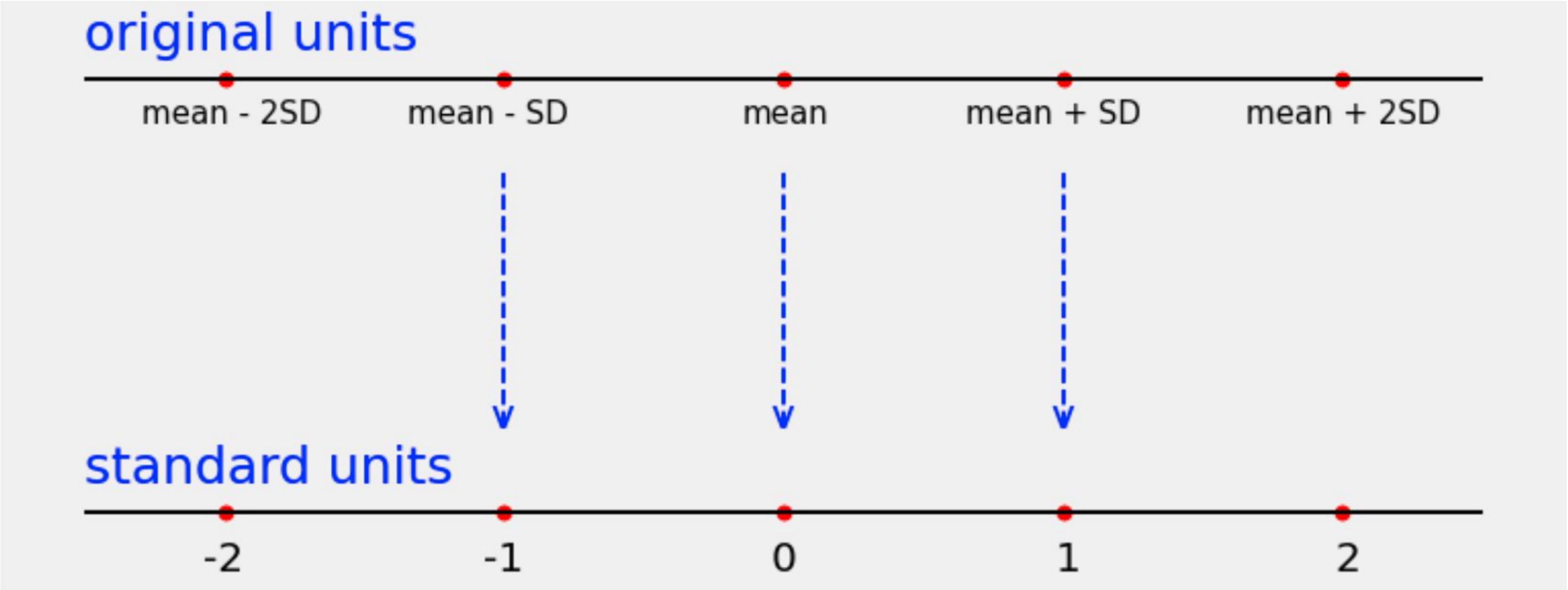
Range	Proportion
average ± 2 SDs	at least 1 - 1/4 = 3/4 (75%)
average ± 3 SDs	at least 1 - 1/9 = 8/9 (88.88…%)
average ± 4 SDs	at least 1 - 1/16 = 15/16 (93.75%)
average ± 5 SDs	at least 1 - 1/25 = 24/25 (96%)

No matter what the distribution looks like!

(Demo)

Standard Units

Standard Units



standard units = (original value - mean) / SD

Standard Units

- Measures: How many SDs above average?
- ***z* = (value - average)/SD**
  - Negative z: value below average
  - Positive z: value above average
  - z = 0: value equal to average

(Demo)

- When values are in standard units: average = 0, SD = 1

Discussion Question

Find whole numbers that are close to:

(a) the average age

(b) the SD of the ages

(Demo)

Age in Years	Age in Standard Units
27	-0.0392546
33	0.992496
28	0.132704
23	-0.727088
25	-0.383171
33	0.992496
23	-0.727088
25	-0.383171
30	0.476621
27	-0.0392546

... (1164 rows omitted)

The SD and the Histogram

- Usually, it's not easy to estimate the SD by looking at a histogram.

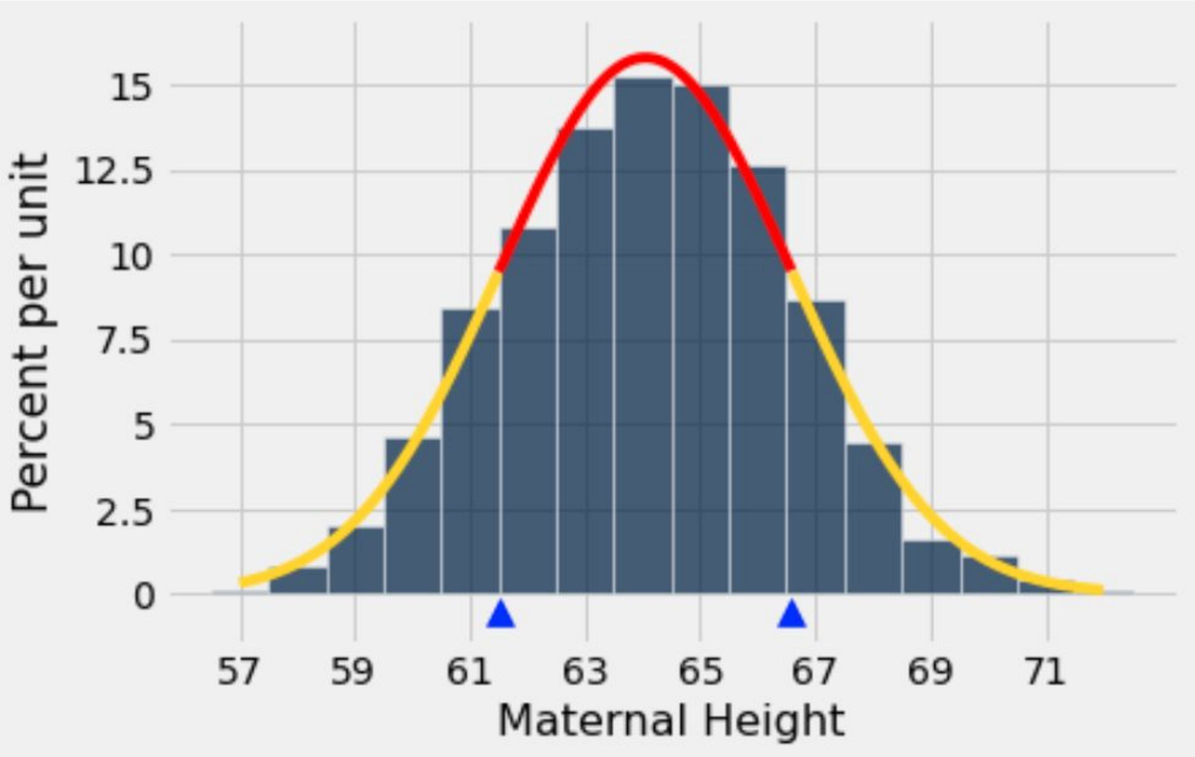
- But if the histogram has a bell shape, then you can.

The SD and Bell-Shaped Curves

If a histogram is bell-shaped, then

- the average is at the center
- the SD is the distance between the average and the points of inflection on either side

Points of Inflection



(Demo)